

Rohan Mulay

+91 7483994847 | rohanm1307@gmail.com | linkedin.com/in/rohan-mulay | github.com/RohanMulay1

SUMMARY

AI Engineer specializing in Computer Vision, ADAS, and real-time perception systems. Experienced in developing and optimizing low-latency deep learning pipelines for real-time inference on edge hardware, from research to production

EDUCATION

PES University

Bachelor of Technology in Computer Science and Engineering

Bengaluru, Karnataka

2022 – 2026

EXPERIENCE

AI Engineer Intern

Cautio (acquired BYTES) – AI-powered dashcam platform

January 2026 – Present

Bengaluru, Karnataka

- Architected ADAS perception features (forward collision warning, lane detection) deployed across 100M+ km, powering 32M+ safety alerts and 719K+ trips across 100+ cities.
- Developed a Driver Monitoring System for facial recognition and drowsiness detection, achieving 96.7% accuracy using a 5K+ embedding database.
- Engineered OCR pipeline for multilingual license plate recognition, achieving 94.2% character-level accuracy.
- Designed and integrated FastAPI microservices with vector search, reducing lookup latency to <50ms at 10K+ queries/day.

AI Engineer Intern

BYTES – ADAS solutions for two-wheeler safety

October 2025 – December 2025

Bengaluru, Karnataka

- Developed an end-to-end ADAS perception pipeline using multi-camera sensor fusion, processing 4 concurrent video streams at 30 FPS on embedded hardware.
- Deployed YOLOv11 on NVIDIA Jetson Orin NX using TensorRT FP16 optimization, reducing latency from 42ms to 11ms while maintaining 91.3% mAP.
- Evaluated FP32, FP16, and INT8 tradeoffs to balance accuracy and real-time inference constraints on edge devices.
- Built and deployed monocular depth estimation pipeline, achieving 8.2% error across 12K+ frames on GCP.

ADAS and Machine Learning Engineer Intern

Moonrider.ai – Autonomous electric tractor startup

April 2025 – October 2025

Bengaluru, Karnataka

- Designed and deployed a semantic segmentation pipeline for drivable area detection, achieving 83.4% mIoU across 6 terrain classes.
- Defined data and validation pipeline with augmentation and LiDAR grounding across 15K+ frames.
- Optimized models using pruning and INT8 quantization, reducing model size by 62% while sustaining 24 FPS on embedded hardware.
- Built LSTM-based models for battery SoC and SoH prediction, achieving 2.1% RMSE across 800+ cycles.

PROJECTS

Integrative Multi-Omic Biomarker Discovery for NAFLD Progression | *PyTorch, GNNs, SHAP*

- Implemented a multi-omic ML framework using Graph Neural Networks and co-attention mechanisms for NAFLD progression prediction, achieving 0.89 AUC-ROC on 230+ patients.
- Integrated SHAP and LIME interpretability modules to surface top-15 discriminative biomarkers for clinical decision support.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Core: Computer Vision, Deep Learning, Sensor Fusion, Edge AI

Frameworks: PyTorch, TensorFlow, OpenCV

Deployment: TensorRT, CUDA, ONNX, NVIDIA Jetson Orin NX, INT8/FP16 Optimization

Backend: FastAPI, REST APIs, Vector Search (Pinecone, Qdrant), MongoDB

Tools: Docker, GCP, Git, Linux, n8n